

Assignment 2 SimQuick Analysis

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I compared the call center with one through five customer service agents under normal demand and increased demand. Each setup had one supervisor and used 100 simulations of an eight-hour day. The tables below include the relevant overall means from my SimQuick results.

Question 1 Normal call volume

Measure	1 agent	2 agents	3 agents	4 agents	5 agents
Service level	85%	95%	97%	97%	96%
Calls unable to enter	24.97	7.49	5.44	5.56	5.88
Average initial wait in minutes	21.43	6.72	3.9	2.83	1.98
Average customers in queue	5.62	2.1	1.25	0.89	0.63
Supervisor time working	61%	72%	74%	74%	74%
Customers finished	124.73	146.63	151.58	148.67	148.36

At 20 calls per hour, I would choose three customer service agents as a reasonable starting point. One agent has a hard time keeping up, with about 25 rejected calls per day and an average initial wait of over 21 minutes. Two agents make a big difference, but the average wait is still almost seven minutes. With three agents, the service level reaches 97% and the average initial wait drops to about four minutes.

Four and five agents reduce the initial wait further, but they do not improve the service level much. The small differences between these results could be from randomness in the simulations. I would not assume that five agents actually make the system worse just because that run had a slightly lower service level.

The utilization results also show why adding agents has limits. With three agents, CS1, CS2, and CS3 are working 45%, 32%, and 20% of the time, but they are blocked about 28-30% of the time. The supervisor is working 74% of the day. This suggests that waiting to pass calls to the supervisor is holding up the regular agents. Their lower working percentages do not mean they are always available for another call.

I think a reasonable goal would be to admit at least 95% of calls and keep the average initial wait below five minutes. Three agents meet both of those goals. Here, service level means calls getting into the system without a busy signal, and the initial wait does not include extra waiting for the supervisor. Before making a final decision, I would want to know staffing costs, the cost of losing calls, customer expectations, and how demand changes during the day. I would also want to look at the longest waits, since an average can hide some bad customer experiences.

Question 2 Increased call volume

For the promotional event, I changed the time between arrivals to Exp(2), which represents about 30 calls per hour. I kept the simulation length at 480 minutes and ran 100 replications again.

Measure	1 agent	2 agents	3 agents	4 agents	5 agents
Service level	58%	76%	80%	81%	81%
Calls unable to enter	100.39	59.43	46.98	44.79	45.65
Average initial wait in minutes	28.61	14.81	9.54	6.85	5.27
Average customers in queue	7.73	5.32	3.63	2.65	2.03
Supervisor time working	64%	86%	90%	90%	90%
Customers finished	128.27	171.5	181.96	185.57	184.29

The increased demand makes the call center perform much worse. With one agent, about 100 calls per day cannot enter, and the average initial wait is almost 29 minutes. Adding a second and third agent helps, but three agents still leave about 47 rejected calls per day and an average initial wait of 9.54 minutes. Compared with normal demand, the three-agent service level drops from 97% to 80%.

Four or five agents reduce the initial wait, but the service level stays around 81%. Even five agents still have an average initial wait above five minutes and about 46 rejected calls per day. None of these setups meets the goals I suggested in Question 1. Four agents finish about 186 customers per day, while five finish about 184, which is close enough that I would not draw a strong conclusion from that difference alone.

The supervisor becomes the main bottleneck. With three agents, the supervisor works 90% of the time. Meanwhile, the agents work 48%, 39%, and 31% of the time and are blocked 44%, 50%, and 53% of the time. Adding more regular agents does not fix this. With five agents, CS5 only works 10% of the day but is blocked 56% of the time.

This also makes sense from the expected workload. If all 30 calls per hour entered, the 30% referral rate would send about nine calls per hour to the supervisor. At eight minutes per call, one supervisor can handle about 7.5 calls per hour on average. That is less than the expected demand. I would test adding another supervisor before hiring more regular agents. Training agents to resolve more issues themselves could also help reduce the number of transfers.

Question 3 Another scenario to explore

One other scenario I would test is three customer service agents with two supervisors during the promotional event. I would keep arrivals at Exp(2), agent call times at Nor(3,2), supervisor call times at Nor(8,3), and the referral rate at 30%. I would also keep the total phone-line limit at 10 so the comparison focuses on the extra supervisor. Calls needing a supervisor would go to whichever supervisor is available.

I would run this setup for 100 eight-hour days and compare it with the three-agent, one-supervisor results from Question 2. I would look at service level, rejected calls, customer waiting, agent blocked time, and how busy both supervisors are. I would expect less blocking, but the simulation would be needed to see how much the results actually improve. Comparing that improvement with the added staffing cost would help decide whether a second supervisor is worth it. This is a proposed scenario, not an additional simulation I ran.

Supporting utilization results

These tables show the working and blocked percentages from my results for all five staffing levels. A dash means that agent was not included in that scenario. Each entry is working time / blocked time.

Normal demand

Employee	1 agent	2 agents	3 agents	4 agents	5 agents
CS1	80% / 16%	53% / 26%	45% / 28%	42% / 29%	42% / 28%
CS2	-	41% / 29%	32% / 29%	28% / 30%	27% / 30%
CS3	-	-	20% / 30%	16% / 30%	15% / 30%
CS4	-	-	-	9% / 29%	8% / 28%
CS5	-	-	-	-	5% / 25%
Supervisor	61% / 0%	72% / 0%	74% / 0%	74% / 0%	74% / 0%

Increased demand

Employee	1 agent	2 agents	3 agents	4 agents	5 agents
CS1	82% / 17%	59% / 38%	48% / 44%	43% / 46%	41% / 47%
CS2	-	52% / 42%	39% / 50%	33% / 51%	31% / 51%
CS3	-	-	31% / 53%	25% / 54%	22% / 54%
CS4	-	-	-	18% / 56%	15% / 56%
CS5	-	-	-	-	10% / 56%
Supervisor	64% / 0%	86% / 0%	90% / 0%	90% / 0%	90% / 0%